

Mapping Mutations to Music: Sonification and Comparative Analysis of *Xylella fastidiosa* Genomic Data

Revised manuscript prepared for student-journal submission

Abstract

Context: Sonification can make sequence patterns audible, but biological usefulness requires validation against known genomic differences rather than subjective listening alone. Objective: This study tested whether a deterministic DNA-to-MIDI encoding preserves measurable variation in *Xylella fastidiosa*, a bacterial plant pathogen associated with Pierce's disease, citrus variegated chlorosis, almond leaf scorch, and olive quick decline syndrome. Methods: Twenty complete RefSeq assemblies were downloaded from NCBI, including grapevine, citrus, coffee, plum, almond, olive, oak, and other host-associated strains. The first 180,000 unambiguous bases of each genome were encoded by a fixed codon-to-pitch map. Audio-derived features included pitch distribution, interval size, pitch entropy, and summary statistics. Validation used 190 pairwise genome comparisons, 300 synthetic mutation trials with known substitution rates from 0.05% to 2.0%, and shuffled-sequence null controls preserving nucleotide composition. Results: Pairwise audio-feature distances were moderately correlated with sequence-feature distances (Pearson $r = 0.441$). A holdout linear model predicted synthetic mutation rate from audio features with $R^2 = 0.916$ and mean absolute error = 0.00127. Shuffled null sequences had a much larger median audio distance from originals (1.308) than synthetic mutation trials (0.0159). Conclusions: Sonification did not replace alignment or variant calling, but it produced reproducible audio features that tracked controlled sequence change. The method is best interpreted as an exploratory and educational display layer for genomic comparison, not as a diagnostic assay.

Keywords: *Xylella fastidiosa*; genome sonification; auditory display; comparative genomics; synthetic mutation benchmark; MIDI.

Introduction

Xylella fastidiosa is not a virus; it is a Gram-negative, xylem-limited bacterium first described by Wells and colleagues [1]. It is transmitted by xylem-feeding insects and causes economically important plant diseases, including Pierce's disease of grapevine, citrus variegated chlorosis, almond leaf scorch, bacterial leaf scorch, and olive quick decline syndrome [2,7,8]. Its biology makes it an appealing case study for comparative genomics: complete genomes are small enough for student-scale analysis, but strains differ by host association, geography, and evolutionary history [3-7].

The original version of this project attempted to compare only three accessions and described the resulting sounds qualitatively. That approach could not support the biological claims it made. Short or incomplete nucleotide records are not equivalent to complete genomes, subjective descriptions such as 'chaotic' or 'harmonic' are not measurements, and BLASTp cannot validate

an audio file. The revised study therefore asks a narrower and testable question: when *X. fastidiosa* DNA is converted to musical events by a fixed rule, do quantitative audio features preserve measurable sequence differences?

Prior studies show that sonification can support exploration and communication of DNA or protein patterns [14-16], and broader auditory-display research provides design principles for mapping data to sound [17,18]. However, a sonification method must be evaluated against ground truth. This study therefore combines real NCBI genomes with synthetic mutations and shuffled null controls. The hypothesis was that audio-feature distances would increase with sequence divergence and would distinguish low-rate synthetic mutation from randomized controls.

Materials and Methods

Genome Selection

Genome metadata were retrieved from the NCBI Datasets API on July 12, 2026. Assemblies were restricted to *X. fastidiosa* records labeled Complete Genome in RefSeq. Twenty assemblies were selected to include classic reference strains and a range of hosts and locations. Table 1 lists all accessions used. The FASTA files and selection metadata are stored locally in `data/selected_genomes.csv` and `data/fastas/`.

Table 1. Complete RefSeq *Xylella fastidiosa* assemblies used in the revised analysis.

Accession	Strain	Host/source	Location	Length (bp)
GCF_011801475.1	ATCC 35879	Grapevine	USA: Florida	2607257
GCF_028891345.1	ICMP 15197	Vitis vinifera	USA: Florida	2600197
GCF_001456235.1	J1a12	Citrus	Brazil: Jales, Sao Paulo	2867237
GCF_001456195.2	3124	Coffee	Brazil: Matao, Sao Paulo	2748592
GCF_001456275.1	U24d	Citrus sinensis	Brazil: Ubarana, Sao Paulo	2732490
GCF_001456295.2	Pr8x	Plum	Brazil: Jarinu, Sao Paulo	2705820
GCF_032594475.1	CFBP8478	Coffea arabica	France: Plant intercepted in from Costa Rica	2676633
GCF_032594455.1	CO33	Coffea arabica	Italy: Plant intercepted in from Costa Rica	2676561
GCF_002954205.1	Salento-2	Olea europaea	Italy: Ugento, Lecce, Apulia	2543566
GCF_042243345.1	CFBP8070	Prunus sp.	USA: Georgia	2867959
GCF_042239055.1	XYL1752/17	Prunus dulcis	Spain: Ciudadella, Menorca	2769601
GCF_055765255.1	CFBP8082	Ambrosia artemisiifolia	USA: Florida	2800184
GCF_012974125.1	RH1	Olea europaea	USA: California	2678425
GCF_012974145.1	LM10	Olea europaea	USA: California	2669650

Accession	Strain	Host/source	Location	Length (bp)
GCF_021459905.1	Oak 35874	Quercus	USA: Washington DC	2648027
GCF_004023395.2	IVIA5901	Prunus dulcis	Spain: Bolulla, Alicante province	2559157
GCF_000019325.1	M12	not reported	not reported	2475130
GCF_000019765.1	M23	not reported	not reported	2573987
GCF_000007245.1	Temecula1	not reported	not reported	2521148
GCF_000006725.1	9a5c	not reported	not reported	2731750

Preprocessing

Each FASTA file was parsed by removing headers and retaining only A, C, G, and T. Ambiguous characters were excluded. To keep the analysis reproducible and computationally reasonable on a student workstation, the first 180,000 clean bases of each genome were used for the benchmark. This standardized segment length avoids giving longer assemblies more feature weight. The limitation is addressed in the Discussion.

Sonification Mapping

DNA was read in non-overlapping codons. The 64 possible codons were ordered lexicographically from AAA through TTT and mapped deterministically to MIDI pitches 36-83 using $\text{pitch} = 36 + (\text{codon index mod } 48)$. Each event had the same duration in the preview MIDI files so that measured differences came from pitch content and pitch intervals rather than performance choices. This intentionally simple map was chosen because it is transparent, repeatable, and easy to audit.

Audio and Sequence Features

Sequence features included nucleotide frequencies, GC proportion, and 64 codon frequencies. Audio features included mean pitch, pitch standard deviation, median pitch, pitch entropy, mean absolute interval size, interval standard deviation, 10th and 90th pitch percentiles, and a 48-bin pitch histogram. Pairwise Euclidean distances were calculated for sequence features and audio features across the 20 genomes.

Validation Design

Three validation layers were used. First, all 190 genome pairs were compared to test whether audio-feature distance tracked sequence-feature distance. Second, synthetic point mutations were introduced at five known rates: 0.05%, 0.1%, 0.5%, 1.0%, and 2.0%. Three replicates per rate per genome produced 300 synthetic mutation records. Third, shuffled null sequences were generated by randomly permuting bases while preserving nucleotide composition. A 75/25 train-test split was used to test whether audio features could predict mutation rate in held-out synthetic examples. All random operations used seed 20260712.

Reproducibility

The pipeline is implemented in Python in `analysis/select_and_download.py` and `analysis/run_benchmark.py`. Outputs include `results/pairwise_distances.csv`, `results/mutation_benchmark.csv`, `results/summary.json`, and MIDI previews in `output/audio/`. The benchmark output hash was `606943f5cf3380a975ef8305be9b01b61da7e5485c384692424ecd7d8d73e30d`.

Results

The analysis used 20 complete RefSeq genomes and 180,000 bases per genome. The 20 assemblies produced 190 pairwise comparisons. Audio-feature distance and sequence-feature distance were moderately correlated (Pearson $r = 0.441$). This result supports the hypothesis that the sonification map preserved some genomic similarity structure, but the correlation was not strong enough to claim that sound alone reconstructs phylogeny or identifies strain type.

In the controlled mutation benchmark, held-out mutation rate was predicted from audio-feature changes with $R^2 = 0.916$ and mean absolute error = 0.00127. The median audio distance for synthetic mutation trials was 0.0159, whereas the median audio distance for shuffled null controls was 1.308. The difference indicates that the audio representation responded to structured sequence changes while treating randomized sequences as far more distant.

The MIDI previews confirmed that the pipeline produces playable sonifications, but the statistical analysis did not depend on human ratings of pleasantness, harmony, or perceived disorder. Listening can help a researcher notice patterns, but the evidence in this study comes from computed features and controlled comparisons.

Discussion

The revised results support a limited claim: deterministic sonification can act as a reproducible display layer that preserves measurable aspects of sequence variation. The study does not show that sonification can diagnose host range, pathogenicity, or subspecies. Those biological conclusions require curated labels, alignments, gene annotation, variant calling, and independent experimental evidence. This distinction is important because *X. fastidiosa* disease outcome depends on host, vector, environment, and bacterial genotype [2,7,8].

The strongest part of the revised design is the ground-truth mutation test. Synthetic substitutions provided known rates that could be compared against audio-feature changes. Shuffled controls provided a negative-control condition that preserved base composition while destroying local order. Together, these controls address the main weakness of the original manuscript: it described sounds but did not test whether the sounds corresponded to known sequence differences.

Several limitations remain. First, the codon-to-pitch map is deterministic but still arbitrary; other musical encodings could emphasize amino acid class, codon degeneracy, GC content, motif

recurrence, or gene boundaries. Second, the first 180,000 bases were analyzed for speed, so the current benchmark is a standardized segment comparison rather than a full-genome alignment. Third, Euclidean distance on summary features is simpler than whole-genome SNP analysis and should not be interpreted as evolutionary distance. Fourth, the dataset was not used to classify disease severity because host labels and pathogenic outcomes are not equivalent.

Future work should compare multiple sonification maps against the same mutation benchmark, include complete genome alignments, use curated subspecies and host-pathogenicity labels, and test whether trained listeners can detect specific sequence changes under blinded conditions. A useful next benchmark would measure sensitivity and specificity for predefined mutations inserted at known positions.

Conclusion

This project was rebuilt from a qualitative demonstration into a reproducible validation study. Across 20 complete *X. fastidiosa* RefSeq genomes, a transparent codon-to-MIDI mapping produced audio features that moderately tracked sequence-feature distances and strongly reflected controlled synthetic mutation rates. Sonification should not be presented as a replacement for standard bioinformatics. Its value is as an interpretable, educational, and exploratory companion to quantitative sequence analysis.

Data and Code Availability

All scripts, selected accession metadata, benchmark outputs, and MIDI preview files are contained in the local project workspace. The main reproducible commands are: `python analysis/select_and_download.py` and `python analysis/run_benchmark.py`.

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